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Fluid Mechanics of Blood Cells and Vesicles Squeezing Through Narrow Constrictions

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Keywords

fluid–structure interactions, soft lubrication, multiscale modeling, Stokesian dynamics, boundary integral methods, erythrocytes, leukocytes, vesicles

Abstract

The squeezing of blood cells and vesicles through narrow constrictions, such as splenic slits, pulmonary capillaries, vascular endothelial gaps, and microfluidic channels, is crucial in physiology and biotechnology, with fluid mechanics playing a central role. The diverse geometries of these constrictions, the associated flow conditions, and the unique mechanical properties of cells and vesicles create a rich subject in fluid mechanics emerging from nonlinear dynamics of fluid–structure interactions involving both lubrication and Marangoni flows. Advances in microfluidics, video microscopy, and computational modeling have enabled investigations into these complex processes. This review surveys the key features and approaches, recent prominent studies, and unresolved challenges related to these processes, offering insights for researchers across biomechanics, biomedical engineering, biological physics, hematology, physiology, and applied mathematics.

1. INTRODUCTION

One of the fundamental challenges of fluid mechanics is to quantify the complex behaviors and properties of particulate flow of red blood cells (RBCs) (erythrocytes), white blood cells (WBCs) (leukocytes), and vesicles under varying conditions in microfluidics, microvascular networks, and cytometry systems, where blood cells and vesicles navigate dynamic environments in single file (Freund 2014, Secomb 2017). It remains an open challenge to understand how, through the intricate coupling between biological and mechanical processes, the microcirculation of blood through the smallest vessels—arterioles, capillaries, and venules—can function with exceptional efficiency and robustness. In this review we focus on the hydrodynamics of deformable particles in a narrow constriction such as blood cells squeezing through microvascular vessels.

While extensive studies have examined deformable particles in constrictions such as tubes and channels, most have focused on cases where the constriction size is comparable to or larger than the particle (Jones 1969, Secomb et al. 1986, Wootton & Ku 1999, Pozrikidis 2003, Hernández-Ortiz et al. 2007, Freund 2014, Freund & Vermot 2014, Barthes-Biesel 2016, Lanotte et al. 2016, Secomb 2017, Beris et al. 2021, Kahali et al. 2024). Here, we consider narrow constrictions, typically less than 1 μm , where giant vesicles and blood cells of several micrometers in diameter must squeeze to pass (Deplaine et al. 2011, Moreau et al. 2023, Mukhopadhyay et al. 2024), giving rise to distinct transport and lubrication dynamics that do not emerge in larger confinements.

Recent advances have been made in our understanding of the mechanisms driving diverse phenomena in blood flow (Barber et al. 2011, Freund & Orescanin 2011, Freund 2014, Henríquez Rivera et al. 2015), including the rheofluidizing properties of blood and the dynamics of cellular aggregation (Kumar & Graham 2012a, 2012b; Beris et al. 2021). In parallel, studies on giant vesicles (synthetic lipid-bilayer membranes as shown in **Figure 1a**) and capsules provide fundamental insights into how blood cells' hydrodynamic and transport characteristics emerge from their structural complexity (Kumar & Graham 2011, Hu et al. 2013, Barthes-Biesel 2016). Much research effort has been devoted to microcirculation disruptions caused by altered blood cell properties (rigidification, increased adhesiveness) and vascular changes (thrombosis) in diseases such as sickle cell disease (Mohandas & Gallagher 2008) and acute respiratory distress syndrome (ARDS), and attention has also been directed at better understanding cell activity, from diapedesis to the activation of mechanosensitive membrane channels (Pak et al. 2015, Mukhopadhyay et al. 2024). Results from these efforts point to the importance of the hydrodynamics of extremely confined cells and vesicles passing through narrow constrictions—down to one-thirtieth of the cell's diameter—relevant to bone marrow, the spleen, pulmonary capillaries, flow cytometers, and microfluidic devices (see **Figure 1**). However, these situations have been little studied, partly because experiments involving constrictions less than 1 μm wide are difficult to perform and partly because modeling is so complex.

The main goal of this review is to elucidate the hydrodynamics of blood cells and vesicles through a narrow constriction, focusing on how their mechanical properties and activity affect their resistance as they squeeze through constrictions of various geometries. We only briefly discuss the chemical and biological interactions between the membrane and the constriction walls.

2. PHYSIOLOGICAL AND MICROFLUIDIC NARROW CONSTRICTIONS

The squeezing hydrodynamics of cells or vesicles is dominated by the soft lubrication flow between a deformable surface (the elastic membrane) and the wall at zero or a low Reynolds number, as cells deform to pass the tapered constrictions under a pressure drop across the passage. In the vascular system such configurations are mainly blood capillaries, which connect arterioles to venules arranged in tree-like networks known as capillary beds, whose walls consist of endothelial cells

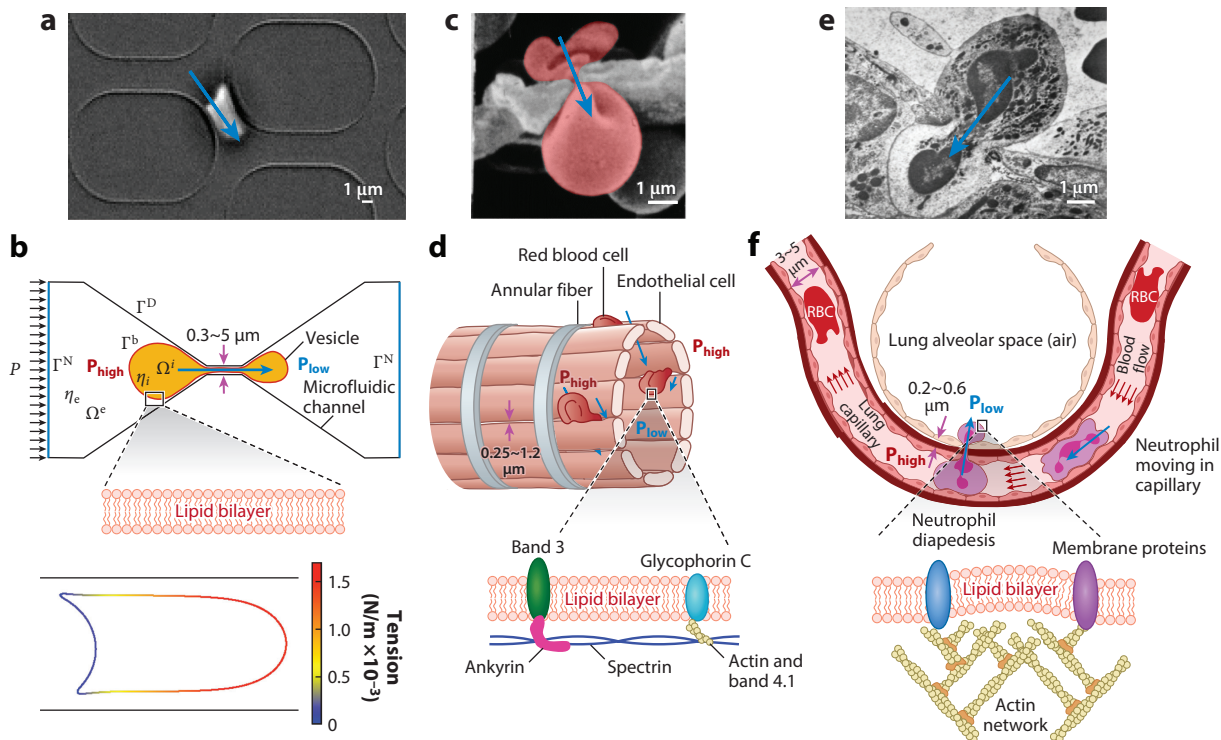


Figure 1

Examples of blood cells and vesicles squeezing through narrow constrictions. (a) A vesicle passing through a microfluidic constriction network. (b) Sketch of a vesicle passing through a pore with the lipid-bilayer structure highlighted, including boundary conditions for fluid mechanics. P is the pressure applied at the inlet, and pressure is zero at the outlet. The external fluid viscosity η_e in Ω^e might be different from the fluid viscosity η_i inside the vesicle Ω^i . The bottom panel shows a vesicle profile, color-coded by membrane tension, in a long channel. Panel adapted from Quafe et al. (2021). (c) Scanning electron microscopy image of a red blood cell (RBC) passing a splenic slit. Panel adapted with permission from Mohandas & Gallagher (2008). (d) Diagram of RBCs passing through the interendothelial slits in the sinus in the spleen with highlighted RBC membrane structures. Panel adapted with permission from Mebius & Kraal (2005). (e) Transmission electron microscopy of a neutrophil passing through the endothelial layer during diapedesis. Panel adapted with permission from Firestein et al. (2016). (f) Sketch of the neutrophils squeezing in the narrow capillary bed in a lung alveolus and diapedesis into the alveolar space. The membrane and actin cortex of the neutrophil are highlighted. All blue arrows in this figure show the moving directions of the vesicles and cells, with purple arrows for dimensions and red arrows for flow directions. P_{high} indicates a high hydrodynamic pressure region and P_{low} indicates a low hydrodynamic pressure region. Panel adapted from images created in BioRender; Peng Z. 2026. <https://BioRender.com/c8d668q>.

coated with a glycocalyx, a brush-like layer of glycoproteins and glycolipids up to 0.5 μm thick and also present on blood cells, acting as a barrier to cell adhesion. The smallest capillaries are normally 3–5 μm in diameter (**Figure 1f**), so that RBCs and WBCs undergo moderate deformation. In systemic capillaries, the flow pushing the cells into the constrictions is generated by a pressure drop of a few thousand pascals (Atkinson & Sherlock 1954), sufficient to deform RBCs and WBCs and allow them easy passage.

Cell deformation is more severe in three physiological situations: The first scenario is the passage of RBCs through submicrometer-wide interendothelial slits (IESs) in the red pulp of the spleen (**Figure 1c,d**), where they must pass a fitness test to stay in the microcirculation. Nascent RBCs (reticulocytes) also pass through narrow slits in the bone marrow, where they are produced, to enter the bloodstream. These slits, formed by the gap between two adjacent endothelial cells,

have a high aspect ratio, extending $5\text{ }\mu\text{m}$ in length, with a depth matching the endothelial thickness ($\sim 2\text{--}3\text{ }\mu\text{m}$) and a variable width ranging from 0.2 to $1.4\text{ }\mu\text{m}$ (**Figure 1d**). The second scenario involves circulating WBCs, particularly polymorphonuclear leukocytes, with neutrophils being the most abundant ($10\text{--}15\text{ }\mu\text{m}$) (Niemic et al. 2015). Their transit through the highly branched alveolar capillary network, where diameters narrow to $2\text{ }\mu\text{m}$ (Doerschuk et al. 1993), presents a significant challenge. The third scenario is diapedesis, the active transmigration of WBCs (primarily neutrophils) between two adjacent endothelial cells that form the capillary or venule wall. Diapedesis begins when a gap forms via the local dissociation of adjacent endothelial junctions (**Figure 1e**). This gap expands to $2.5\text{--}3\text{ }\mu\text{m}$ during diapedesis before the junctions reseal (Yeh et al. 2018). Unlike pressure-driven squeezing, diapedesis is an active process guided by inflammatory signals and intricate cell–cell interactions. Active morphological remodeling of the membrane and cytoskeleton in both endothelial cells and leukocytes is required for diapedesis, which unfolds over several minutes.

In synthetic devices, polycarbonate sieves with uniform cylindrical pores (2 and $6\text{ }\mu\text{m}$ in diameter) have been used to filter RBCs and investigate their mechanical properties (Gregersen et al. 1967). Many newly developed microfluidic devices with different constriction geometries (at least $2\text{ }\mu\text{m}$ in channel width) and successions of several constrictions were designed to assess RBC deformability (Matthews et al. 2022, and references therein) or were dedicated to cell analysis, sorting, and isolation (Gregersen et al. 1967, Yu et al. 2014, Kabacaoğlu & Biros 2019, Farahinia et al. 2021). Yet the extent of cell deformation is limited by the minimal channel width of $2\text{ }\mu\text{m}$ and by the conventional soft optical microlithography techniques. Various approaches, adapted to the aspect ratio of the constrictions, have been explored: One method combines optical lithography with asymmetrical chemical etching to create slits comparable to interendothelial structures (Gambhire et al. 2017, Garcia-Herreros et al. 2022, Moreau et al. 2023). Another method involves stacking microspheres to form confined pathways, offering high-throughput filtration but without real-time visualization (Tandogan et al. 2014, Duez et al. 2018).

Finally we note that, to our knowledge, there is no experimental work on giant vesicles squeezing through such submicrometer constrictions. This is because only vesicles with large excess areas (highly flaccid for large deformations) could pass through, but they are difficult to manipulate as they pearl, bud, and split under a moderate pressure drop. However, vesicle hydrodynamics in confinement has been studied both theoretically and numerically (Barakat & Shaqfeh 2018, Quaipe et al. 2021, Gannon et al. 2024). Vesicles reconstituted with actins (Bashirzadeh et al. 2021, Bashirzadeh & Liu 2022) are possible synthetic candidates to squeeze through submicrometer constrictions.

3. STRUCTURES AND MECHANICS OF VESICLES, RED BLOOD CELLS, AND WHITE BLOOD CELLS

In this section we provide a brief survey of structures and mechanical properties of vesicles and blood cells, focusing on microscopic structures of vesicles/cells that give rise to distinct macroscopic mechanical properties.

3.1. Vesicles

A giant lipid vesicle is a fluid sack enclosed by a lipid-bilayer membrane of radius on the order of 5 to $50\text{ }\mu\text{m}$. Unilamellar fluid-phase lipid bilayers behave as 2D incompressible fluids, displaying elastic responses under external stress (**Figure 1b**). Their elastic properties, which vary across different lipid types, are well-documented in the literature. In general, the lipid-bilayer membranes lack in-plane shear resistance and present a small 2D shear viscosity, approximately 10^{-9} – $10^{-8}\text{ Pa}\cdot\text{s}\cdot\text{m}$, and a negligible shear elasticity (Luisi & Walde 2008, Dimova & Marques 2019).

Lipid-bilayer membranes are often assumed to be inextensible, as they have strong resistance to in-plane dilatation; hence the change in (both local and global) surface area is constrained to be small. Beyond in-plane constraints, lipid-bilayer membranes resist out-of-plane bending, twisting, lipid tail tilting, and saddle-splay deformations in their mechanical responses commonly described by the Helfrich energy. Spontaneous curvature and additional moduli (Nagle & Tristram-Nagle 2000), some of which are traced to the collective behaviors of amphiphilic macromolecules in a polar viscous solvent (Fu et al. 2022), can be incorporated into the extended Helfrich membrane energy. Under high stress the lipid membrane porates and ruptures as its tension exceeds the lysis threshold.

Lipid-bilayer membranes may consist of multiple lipid species in each leaflet. The inhomogeneity in lipid species gives rise to a heterogeneous bending stiffness closely related to wrinkling and budding of the membrane. Finally, lipid-bilayer membranes are permeable to water molecules, with a water permeability coefficient k_w proportional to the osmotic water permeability Ω_w : $k_w = \Omega_w/(RT)$ ms^{-1}/Pa , R is the ideal gas constant, and T is the temperature in Kelvin. For a lipid-bilayer membrane, we find the water permeability $k_w \sim 10^{-14}$ ms^{-1}/Pa (Reeves & Dowben 1970) while for a cell membrane $k_w \sim 10^{-11}$ ms^{-1}/Pa , depending on the density of aquaporins in the membrane (Li et al. 2015). In the absence of aquaporins or membrane pores induced by external stretching, the vesicle membrane is typically treated as impermeable, such that its configuration is fully determined by its constant surface area A and volume V . Under this assumption, deformation is only possible for a nonspherical vesicle with a reduced volume $v \equiv 6\sqrt{\pi}V/A^{3/2} < 1$.

3.2. Red Blood Cells

The structure of RBCs has been thoroughly studied and well-characterized (Mohandas & Gallagher 2008, Lux 2016), yet recent advancements in proteomics and imaging techniques provided additional insights that have evolved the mechanical model of an RBC (Feng et al. 2020). At a basic level, a human RBC is an elastic biconcave discoid bag filled with a hemoglobin solution, with its shell consisting of two primary layers: the lipid/protein bilayer, mechanically close to that of vesicles, and a nearly 2D spectrin cytoskeleton anchored in the lipid bilayer (**Figure 1d**). The lipid bilayer ensures a constant surface area, as in vesicles. The spectrin network provides a shear elasticity that is responsible for elastic buckling of the RBC membrane. Numerous studies indicate that the resting biconcave RBC shape arises from a prestressed buckling state, with the membrane's stress-free configuration being closer to a sphere than a biconcave disc (Lim et al. 2002). This intrinsic prestrain plays a crucial role in governing RBC deformations under hydrodynamic stress. Spectrin domain unfolding and dissociation (Zhu & Asaro 2008, Moreau et al. 2023) contribute to the three deformation regimes of shear elasticity: A small deformation regime manifests as the initial shear modulus, closely tied to the resting biconcave shape and stress-free state (Peng et al. 2014). In the intermediate regime, strain-hardening behavior is observed, likely due to the transition from entropy-dominant to enthalpy-dominant behaviors. In the large deformation regime, the unfolding of spectrin domains leads to strain-softening behavior (Zhu & Asaro 2008).

Membrane proteins constitute nearly half the mass of the RBC bilayer membrane and play essential roles in its behavior. For example, proteins anchoring the spectrin-actin cytoskeleton to the lipid bilayer remain mobile within the membrane, allowing the spectrin network to deform locally and independently of the bilayer. Meanwhile, integral membrane proteins, such as ion channels and aquaporins, regulate ion exchange across the membrane, modulating cytoplasmic osmolarity and, consequently, RBC volume and water content. Notably, the mechanosensitive PIEZO1 cation channel might respond to mechanical stress on the membrane (Cahalan et al.

2015), enabling the cell to actively adjust its volume, although there is still a debate on its role in RBCs (Kaestner et al. 2025, Moreau et al. 2025). The activation times of these channels vary, ranging from milliseconds for PIEZO1 to seconds or longer for ion pumps.

The hemoglobin solution within RBCs can be reasonably modeled as a Newtonian fluid with a viscosity several times greater than that of plasma. Hemoglobin exhibits strong sensitivity to both concentration and temperature, undergoing phase transitions (Kelemen et al. 2001) that can be modulated by changes in RBC hydration, including dehydration and overhydration (Moreau et al. 2023).

3.3. Leukocytes (White Blood Cells)

Leukocytes, or WBCs, are immune cells responsible for defending the body against infections and foreign invaders (Muller 2013). Much more complex than vesicles and RBCs, leukocytes are characterized by the presence of a nucleus (**Figure 1e**) and actomyosin machinery of active processes to generate force (**Figure 1f**) and alter their morphologies to achieve motility and immune responses such as engulfing and digesting bacteria.

As the body's primary responders to microbial threats, neutrophils constitute 60–70% of circulating WBCs and have a lifespan ranging from 5 to 135 h in the bloodstream (Amulic et al. 2012). As the most studied WBCs, neutrophils are characterized by their multilobed nuclei, which give them a polymorphonuclear appearance. Unlike RBCs, their bilayer cytoskeletal membrane is extensible due to the presence of extra membrane stored in microvilli that can be released on demand (Shao et al. 1998).

Besides a multilobed nucleus, the neutrophil has an active actomyosin network (**Figure 1f**) that is more dynamic than the passive spectrin network in RBCs (**Figure 1d**). In addition, its intracellular rheology is more controversial and complicated than the hemoglobin solution in RBCs. For example, earlier works (Evans 1989, Needham & Hochmuth 1990) showed that, under a constant suction pressure in a micropipette, neutrophils exhibit behavior akin to that of a Newtonian liquid drop, characterized by cortical tension, when in a resting state. The cytoplasmic viscosity of resting neutrophils and other white cells was estimated as 10^2 Pa·s, a value approximately 10^5 times higher than that of an RBC ($\sim 10^{-3}$ Pa·s). However, later experiments showed that the cytoplasm in neutrophils exhibits behaviors ranging from a Maxwell viscoelastic fluid to a non-Newtonian power-law fluid (Dong et al. 1988, Sung et al. 1988, Tsai et al. 1993, Shirai 2008, Khismatullin 2009). Since the neutrophil has a 3D cytoskeleton, a nucleus, and other organelles, the rheological properties of such an active cytoskeletal gel may not be captured by a simple empirical model of materials, and more sophisticated microstructure-based models are needed, as reviewed by Mogilner & Manhart (2018). In terms of computational modeling, complicated integrations between different components such as cytoskeletal dynamics, cytosol flow, adhesion dynamics (King & Hammer 2001), signaling pathways, and membrane mechanics need to be carefully incorporated to capture the mechanisms of cell migration of neutrophils and other cells (Holmes & Edelstein-Keshet 2012). One example focusing on neutrophil mechanics is the continuum–mechanical model of the neutrophil, which includes active force production from a cytoskeletal swelling force and a polymerization force (Herant et al. 2003).

4. THEORETICAL AND COMPUTATIONAL APPROACHES

In Section 4.1, we review a continuum modeling framework for fluid–structure interactions of an elastic, area-incompressible membrane in a Stokesian fluid within a tapered channel. This framework yields a system of partial differential equations with nonlinear boundary conditions, solvable via boundary integral (BI) (Section 4.2) or particle-based methods (Section 4.3).

4.1. Theory of Vesicles and Viscous Drops in Narrow Constrictions

We consider a vesicle or an RBC inside a tapered constriction as illustrated in **Figure 1b**. Under most physiological conditions, the fluid-phase lipid-bilayer membrane of a vesicle and an RBC can be regarded as a 2D incompressible fluid (with a Helfrich elastic free energy for the surface Γ^b and $\mathbf{n}$ its outward unit normal) that couples to the viscous bulk fluids of constant viscosities η_e in the exterior domain Ω^e and η_i in the interior domain Ω^i . The governing equations for the two bulk fluids are

$$\text{In } \Omega^k : -\eta_k \Delta \mathbf{U}_k + \nabla P_k = -\nabla \cdot \boldsymbol{\tau}_k = \mathbf{0}; \quad \boldsymbol{\tau}_k = -P_k \mathbf{I} + 2\eta_k \mathbf{E}_k; \quad \nabla \cdot \mathbf{U}_k = 0, \quad 1.$$

where $\mathbf{E}_k = \frac{1}{2}(\nabla \mathbf{U}_k + \nabla \mathbf{U}_k^T)$ is the symmetric velocity gradient tensor, with subscript $k = i(e)$ for the interior (exterior) fluid. On the channel wall Γ^D , the fluid velocity $\mathbf{U}_e$ equals zero. On the membrane Γ^b , the in-plane flow field $\mathbf{v}$ is incompressible due to the inextensibility of the lipid-bilayer membrane and satisfies the in-plane stress balance

$$(\boldsymbol{\tau}_e|_{\Gamma^b} - \boldsymbol{\tau}_i|_{\Gamma^b}) \cdot \mathbf{n} = \mathbf{f}_{\text{mem}}, \quad \nabla_s \cdot \mathbf{v} = 0, \quad 2.$$

where $\mathbf{f}_{\text{mem}} = -\kappa(4H^3 - 4KH + 2\nabla_s^2 H)\mathbf{n} + 2\sigma H\mathbf{n} - \nabla_s \sigma$ is the traction on the area-incompressible membrane with H the mean curvature (trace of the curvature tensor on Γ^b) and K the Gaussian curvature. σ is the membrane tension that enforces the area incompressibility and $\nabla_s \sigma$ is its surface gradient on Γ^b . For a lipid-bilayer membrane, the bending rigidity $\kappa \sim 10k_B T$. For an RBC, $\mathbf{f}_{\text{mem}}$ has a substantial contribution from the cytoskeletal spectrin network described in Section 4.2.3. Various particle-based coarse-grained models of lipid-bilayer membranes have been developed to simulate vesicle dynamics (Cooke et al. 2005; Noguchi & Gompper 2005a; Yuan et al. 2010; Fu et al. 2017, 2022) and RBC dynamics (Noguchi & Gompper 2005b, McWhirter et al. 2009, Islam et al. 2023).

Vesicle hydrodynamics in unbound space has been well-studied (Abreu et al. 2014, Steinberg & Levant 2019), while hydrodynamics of vesicles under strong confinement (in microfluidic channels or capillaries; see **Figure 1b**) is relatively less investigated due to the challenge of resolving the complex coupling between the vesicle membrane, fluid flow, and confinement geometries. Vitkova et al. (2004) reported experimental observations of left–right asymmetric vesicle shapes in a channel flow. Trozzo et al. (2015) used numerical simulations to reproduce the asymmetric vesicle shapes and also uncovered several vesicle shapes not observed in the experiments. Central to the vesicle hydrodynamics under strong confinement, closely fit in a channel of constant cross section (Barakat & Shaqfeh 2018, Rallabandi et al. 2021) or contracting geometry (Quaife et al. 2021, Gannon et al. 2024), is the lubrication flow in the thin film of viscous fluid between the deformed membrane and the solid wall, similar to the soft lubrication flow between a rigid particle and an elastic wall (Skotheim & Mahadevan 2004; Rallabandi et al. 2018, 2021; Rallabandi 2024).

The steady shape and speed of a vesicle going through a closely fit channel of constant cross section have been theoretically investigated using a matched asymptotic calculation of the lubrication flow between the vesicle membrane and the confinement wall (Secomb 2017, Barakat & Shaqfeh 2018). These calculations showed that, for a given channel radius, the vesicle reduced volume ν must be sufficiently small for the vesicle to flow through the closely fit channel without lysing, in good agreement with experiments with swollen RBCs (Savin et al. 2016). Below the critical radius ratio, the vesicle reaches a steady equilibrium shape that translates with constant speed and pressure drop, both determined solely by the ratio of the effective vesicle radius to the tube radius. For a vesicle membrane moving in a tapered constriction or a pore of finite length, such matched asymptotic calculation is challenging, as the vesicle shape is highly dynamic during the squeezing process. Numerical simulations of a vesicle in a channel with a tapering cross section (Quaife et al. 2021, Gannon et al. 2024) show that the thickness of the lubrication layer

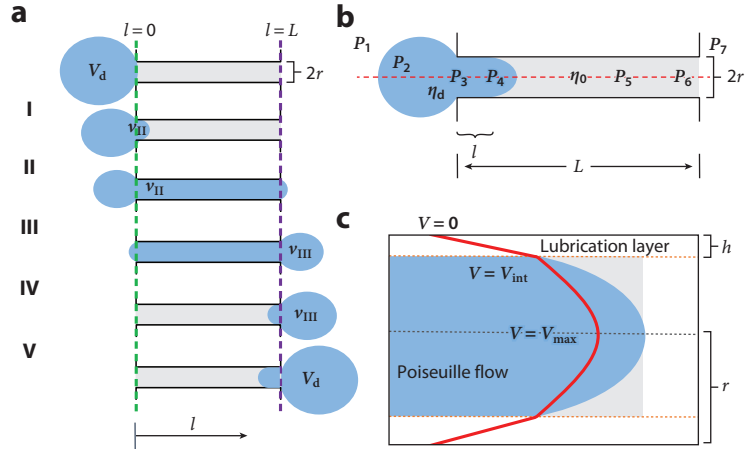


Figure 2

Sketch of a viscous drop of volume V_d squeezing through a cylindrical pore of radius r and length L . (a) Stages of the drop's passage through the pore, denoting v_{II} as the drop volume to the right of $l = 0$ (green dashed line) and v_{III} as the drop volume to the right of $l = L$ (purple dashed line). (b) Dimensions of the studied system (pore and drop dimensions), viscosities η_d and η_0 of the drop and external fluid, respectively, and pressures (P_1 to P_7) outside and along the drop/pore. (c) Velocity profile inside the pore with a lubrication layer between the drop and the pore wall. Figure adapted from Tang et al. (2023).

varies drastically when the vesicle reaches the neck. The flow within this thin layer can be decomposed into contributions from shear-driven motion (Couette flow, linear in the gap height) and pressure-driven motion (Poiseuille flow, quadratic in height). Such dynamic lubrication flow is further complicated when the wall is deformable, giving rise to an effective repulsion from the wall (Beaucourt et al. 2004, Zhang et al. 2020).

Tang et al. (2023) investigated the dynamics of a viscous drop transit through a cylindrical pore as a model to elucidate the dynamics of cell squeezing through narrow constrictions. **Figure 2a** depicts stages of a viscous drop passing through a pore under a pressure drop $\Delta P \equiv P_1 - P_7$ with simple shapes at each stage. We denote v_{II} as the drop volume to the right of $l = 0$ and v_{III} as the drop volume to the right of $l = L$. Stage I is a short process of developing a hemisphere at the pore entrance with a radius of r . Stage II starts when $v_{II} > 0$ and ends when the volume $v_{II} = L\pi r^2 + 2/3\pi r^3$. Stage III ends when $v_{II} = V_d - 2/3\pi r^3$, where V_d is the drop volume. Stage IV ends when $v_{III} = V_d - 2/3\pi r^3$. Stage V is the brief process of the drop at the exit expanding into a spherical drop. Due to the finite length of the pore, the pressure distribution is a function of the drop position as labeled in **Figure 2b**. In this model, velocity and the drop surface profiles inside the pore are assumed to be fixed with a lubrication layer between the drop and the pore wall (see **Figure 2c**). The overall pressure drop across the pore is expressed as the sum of four components:

$$\begin{aligned} \Delta P &= \Delta P_{\text{mem}} + \Delta P_{\text{Poise}, \eta_d} + \Delta P_{\text{Poise}, \eta_0} + \Delta P_{\text{Sampson}} \\ &= 2\sigma \left(\frac{1}{R_R} - \frac{1}{R_L} \right) + \frac{(V_{\text{max}} - V_{\text{int}})4\eta_d l}{(r - b)^2} + \frac{8Q\eta_0(L - l)}{\pi r^4} + \frac{3Q(\eta_0 + \eta_d)}{2r^3}, \end{aligned} \quad 3.$$

where the right-hand side of Equation 3 shows the pressure drops due to membrane tension σ (Young–Laplace equation), Poiseuille flow inside the drop, Poiseuille flow outside the drop, and Sampson flow for the entrance and exit (Sampson 1891), respectively. l is the length of drop inside

the pore, $V_{\max}$ is the maximum velocity and V_{int} is the velocity of the interface between the drop and the lubrication layer, b is the thickness of the lubrication layer, η_d is the drop viscosity, η_0 is the exterior fluid viscosity, and Q is the flow rate as shown in **Figure 2c**. In the limit of $\Delta P_{\text{mem}} \ll \min(\Delta P_{\text{Poise}, \eta_d}, \Delta P_{\text{Poise}, \eta_0})$, the normalized dimensionless transit time $\bar{t}_T = \frac{t_T \Delta P}{\eta_d}$ for the viscous drop is computed as

$$\bar{t}_T = 8(1 + \lambda)\bar{L}^2 + 3\pi(1 + \lambda)\bar{L} + \left(\frac{4\pi}{3}\bar{R}_d^3 - \pi\bar{L} - \frac{4\pi}{3}\right)\left(3 + \frac{8}{\pi}\bar{L}\right), \quad 4.$$

where $\lambda = \eta_0/\eta_d$, $\bar{L} = L/r$, and $\bar{R}_d = R_d/r$ with $R_d = (3V_d/4\pi)^{1/3}$.

For a membrane pressure drop ΔP_{mem} comparable to $\min(\Delta P_{\text{Poise}, \eta_d}, \Delta P_{\text{Poise}, \eta_0})$, Tang et al. (2023) conducted computational investigations into how dimensionless parameters, such as the normalized transit time $\bar{t}_T$, vary with geometric ratios $\bar{L}$ and $\bar{R}_d$ and the capillary number $\text{Ca} = \eta_d V_{\max}/\sigma$. In this model, the minimum pressure drop needed for a viscous drop of a given surface tension to squeeze through a cylindrical pore is computed as $\Delta P_c = 2\sigma\{1/r - [(V_d - 2\pi r^3/3)/(4\pi/3)]^{-1/3}\}$, and the transit time depends on the viscosity ratio λ (Moreau et al. 2023, Tang et al. 2023). A follow-up study explored the analytical and numerical modeling of drop transit through rectangular slits under pressure-driven Stokes flows (Borbas et al. 2023).

Similar to the case of a closely fit vesicle in an infinitely long cylindrical channel (Barakat & Shaqfeh 2018), ΔP_c is a function of geometries of the channel (pore radius) and the drop (volume). However, a vesicle membrane (of a constant surface area) ruptures when its tension corresponding to ΔP_c for passage is above lysis tension (Barakat & Shaqfeh 2018). Consequently, only vesicles with a larger surface area (and hence lower minimum tension for passage) can squeeze through. For tapered narrow constrictions, the size of the constriction neck and the confinement gradient determine the condition for the vesicle's passage in numerical simulations (Quaife et al. 2021), but no matched asymptotic calculations for a vesicle membrane in such constriction geometries are available in the literature.

For blood cells, more dimensionless numbers arise from the membrane bending rigidity, shear elasticity, and area elasticity (Pozrikidis 2003). The dependencies of the detailed squeezing dynamics (such as the transit time t_T) on constriction geometries, membrane mechanics, and bulk mechanics make it difficult to use dimensionless groups to obtain a universal characterization of fluid–structure interactions in the squeezing flow of vesicles and cells, as in the case of cancer cells (Byun et al. 2013, Khan et al. 2017).

4.2. Continuum-Based Computational Models and Boundary Integral Methods

The squeezing hydrodynamics of vesicles and blood cells in narrow constrictions has been numerically simulated by two main groups of methods. The first group is BI methods, which are well-studied for highly accurate simulations of fluid–structure interactions. BI methods offer high computational efficiency for simulating Stokes flow in both free-space and strongly confined geometries coupled to deformable membranes (Freund 2013, Salehyar & Zhu 2016, Lu & Peng 2019). The membrane is typically represented using mesh-based or spectral-element discretizations, with formulation either based on the membrane force–balance equations (Pozrikidis 2003) or derived from the principle of virtual work (Peng et al. 2011). The second group is particle methods such as dissipative particle dynamics (DPD) (Peng et al. 2013), smoothed hydrodynamic particle method (Hosseini & Feng 2012), and multiparticle collision dynamics (Noguchi & Gompper 2005b, McWhirter et al. 2009), where vesicles and RBC membranes are modeled as triangular networks of particles with bond, angle, and dihedral interactions (Discher et al. 1998).

The immersed-boundary (IB) method, popular in simulating fluid–structure interactions, is less used for simulating the squeezing hydrodynamics of cells through extremely narrow constrictions. In this method, the fluid domain is discretized using either the finite difference method (Yazdani & Bagchi 2011, Balogh & Bagchi 2017) or the lattice Boltzmann method (Sui et al. 2008), and the membrane is discretized using the finite element method. The forces from membrane nodes are distributed to the fluid domain via regularized Dirac delta functions. IB methods have lower accuracy in capturing the membrane dynamics, and a very fine mesh is required to resolve the soft lubrication flow between the membrane and the walls. In addition, phase-field methods (Du et al. 2004, Shen et al. 2022) and the arbitrary Lagrangian–Eulerian method using the finite element method (Barber et al. 2011) have also been used for simulating vesicles and cells in a channel.

4.2.1. Boundary integral methods for a constricted space under prescribed inlet and outlet pressures. BI methods are widely used for simulations of vesicles and blood cells with periodic (Zhao et al. 2010) or pure Dirichlet (Agarwal & Biros 2020, Quaife et al. 2021, Gannon et al. 2024) boundary conditions. However, in both experimental settings and physiological contexts where vesicles or RBCs are driven through a constriction by an applied pressure, a Neumann boundary condition (prescribed pressure) is more appropriate than a Dirichlet condition. While previous work has addressed this pressure-driven configuration under the simplifying assumption that the inlet and outlet lie far from evolving interfaces (Quaife & Moore 2018), here we provide a more general formulation.

In continuum-based models, the flow velocity in Ω^e and Ω^i satisfies the incompressible Stokes equations in Equation 1. Equation 2 gives the boundary condition on the elastic membrane Γ^b (Figure 1b). We denote the wall boundary as Γ^D and the inlet/outlet Neumann boundary as Γ^N . On Γ^D , the flow velocity $\mathbf{u}$ is prescribed as zero for fixed walls, but can be specific values for a prescribed velocity. The traction $\mathbf{f}^{(D)}$ on Γ^D is unknown and needs to be determined. On Γ^N , the traction including pressure is prescribed, but the velocity $\mathbf{u}^{(N)}$ on Γ^N is unknown and needs to be determined. On the membrane Γ^b , the traction discontinuity is calculated from the membrane mechanics based on the membrane deformation using the finite element method, while the velocity $\mathbf{u}^{(b)}$ is unknown. The system of equations for $\mathbf{f}^{(D)}$, $\mathbf{u}^{(N)}$, and $\mathbf{u}^{(b)}$ (see the **Supplemental Appendix**) is then solved using iterative schemes.

4.2.2. Cell–wall interactions under strong confinement: singular integrals, strong lubrication, and repulsive forces. Simulating the passage of an RBC through a narrow slit in the BI framework requires precise evaluation of singular and nearly singular integrals to resolve the strong hydrodynamic interactions between the cell and the channel walls. Specialized quadrature methods are employed to compute such near-singular integrals (Quaife & Biros 2014), and explicit avoidance of contact between membranes and rigid boundaries can be incorporated in the BI formulation to reduce the stiffness for resolving near-contact hydrodynamics (Lu et al. 2017), particularly when the RBC membrane approaches within 50 nm of the wall or when the cell surface nearly contacts itself (Zhu et al. 2017). For example, a self-adaptive quadrature method (Lu & Peng 2019) was developed to address these computational challenges effectively, simplifying the numerical framework while ensuring accuracy in capturing the pronounced lubrication hydrodynamics.

In addition, a repulsive force is applied to account for the glycocalyx layer’s steric and electrostatic effects to model the interaction between an RBC membrane and the channel wall. The glycocalyx layer, which is 5–10 nm thick on the RBC membrane and even thicker on endothelial cells, is negatively charged, as are the channel walls, causing a significant repulsive force when the cell–wall distance is below 35 nm. This repulsive force is modeled as a harmonic force and added to the membrane force in the BI formulation (see the **Supplemental Appendix**).

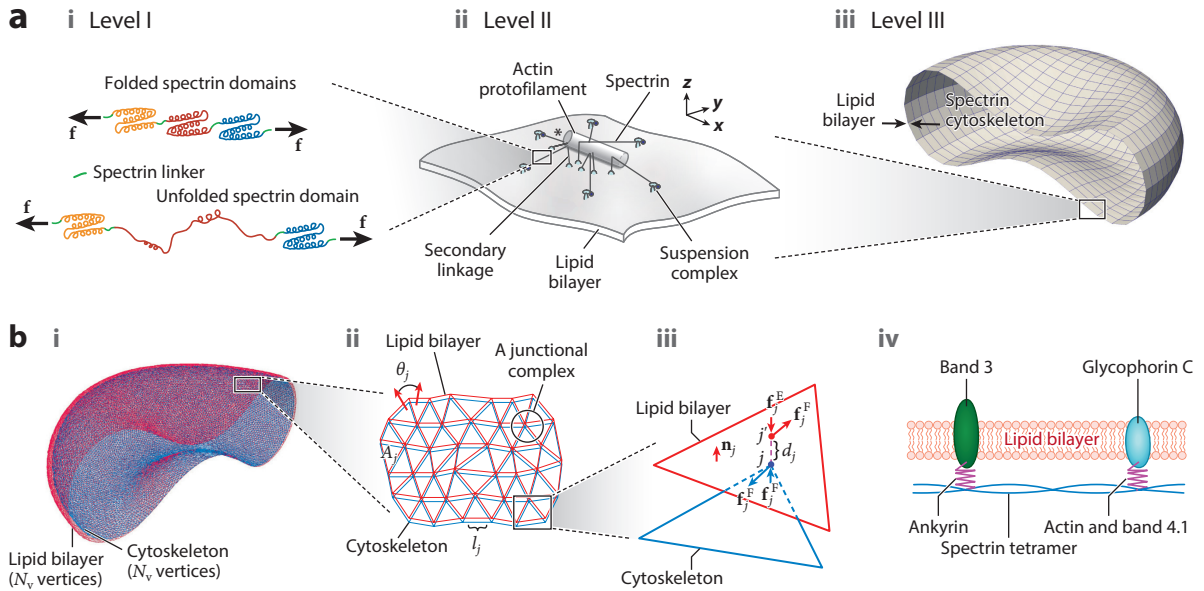


Figure 3

Two multiscale models of the red blood cell (RBC) membrane. (a) Three-level multiscale model of an RBC. (a, i) Level I, spectrin constitutive model. (a, ii) Level II, junctional complex model. (a, iii) Level III, whole-cell model simulated by the finite element method with the bilayer and cytoskeleton modeled as two distinct layers interacting through a penalty contact algorithm. Panel a adapted from Lu & Peng (2019). (b) Two-component dissipative particle dynamics model of the whole cell. (b, i) Lipid bilayer (red) and the cytoskeleton (blue) as two distinct components. (b, ii) Local triangular networks of the two-component model. l_j is the spring length of the cytoskeleton, θ_j is the instantaneous angle between two adjacent triangles on the bilayer, and A_j is the triangle area. (b, iii) Normal and tangential interactions between the lipid bilayer and the cytoskeleton. $\tilde{j}$ is the projection point on the lipid bilayer of vertex j on the cytoskeleton, d_j is the distance between point j and point $\tilde{j}$, and $\mathbf{f}_j^F$ is the tangential friction interaction force, while $\mathbf{f}_j^E$ is the normal elastic interaction force and $\mathbf{n}_j$ is the normal direction vector of the bilayer triangle. (b, iv) Physical picture of the local bilayer-cytoskeletal interaction. Each junctional complex involves two categories of interactions: major linkages through band 3 and ankyrin and secondary connections mediated by actin, glycophorin C, and band 4.1. These are treated collectively as an effective bilayer-cytoskeletal coupling, modeled as a normal elastic force combined with a tangential frictional force. For clarity, the vertical damping force $\mathbf{f}_j^D$ and the random force $\mathbf{f}_j^R$ are not depicted in the figure. Panel b adapted from Peng et al. (2013).

4.2.3. Multiscale modeling of red blood cell membranes and spectrin unfolding. The elastic membrane mechanics of RBCs can be described through empirical or multiscale constitutive laws that capture mechanical response across different length scales. In a three-level multiscale framework (Peng et al. 2010, Feng et al. 2020, Moreau et al. 2023), the Cauchy stress resultant at the whole-cell continuum level (Figure 3a, subpanel iii) is expressed as

$$\boldsymbol{\sigma}^s(\mathbf{F})\mathbf{b} = \tau_m(\alpha, \beta)\mathbf{I} + \frac{\mu(\alpha, \beta)}{(\alpha + 1)^2} \left(\mathbf{B} - \frac{\text{trace}(\mathbf{B})}{2}\mathbf{I} \right), \quad \mathbf{B} = \mathbf{F}\mathbf{F}^T, \quad 5.$$

with $\mathbf{b}$ the thickness of the membrane modeled as a thin shell and τ_m and μ the membrane mean stress resultant and shear modulus, respectively. $\mathbf{B}$ is the left Cauchy–Green deformation tensor, $\mathbf{F}$ is the deformation gradient tensor, and $\mathbf{I}$ is the identity matrix. $\alpha = \lambda_1\lambda_2 - 1$ and $\beta = (\lambda_1^2 + \lambda_2^2)/(2\lambda_1\lambda_2) - 1$ are two invariants representing the area and shear deformation, respectively (Vaugh & Evans 1979). λ_1 and λ_2 are principal stretches calculated from $\mathbf{F}$. Details of the multiscale model, including the expressions of τ_m and μ and the junctional complex model of

the cytoskeleton in **Figure 3a**, subpanel *ii*, and spectrin unfolding in **Figure 3a**, subpanel *i*, can be found in the **Supplemental Appendix**.

4.3. Particle-Based Computational Models and Dissipative Particle Dynamics

The DPD method is a coarse-grained numerical scheme widely applied to simulate soft colloidal suspensions (Winkler et al. 2014, Xu et al. 2019), and it has been successfully applied to study RBCs in flows (Fedosov et al. 2010; Peng et al. 2013, 2016; Pivkin et al. 2016; Perazzo et al. 2022). A DPD system is represented by N particles, which interact through pairwise effective potentials and move according to Newton's second law (Hoogerbrugge & Koelman 1992, Groot & Warren 1997). In a DPD simulation, a particle represents the center of mass in a cluster of atoms or molecules, and the position and momentum of the particle are updated in a continuous phase but spaced at discrete time steps.

In the DPD method, the dissipative force and a random force act as a heat sink and source, respectively, and the combined effect of the two forces acts as a thermostat. Also, a common choice of the soft repulsion for the conservative force permits the usage of larger integration time steps than usually allowed in the molecular dynamics simulation, making DPD a simple and efficient method for simulating hydrodynamic interactions. Peng et al. (2013) and Pivkin et al. (2016) developed a two-component DPD model for the RBC membrane (for details, see **Figure 3b**; **Supplemental Appendix**).

The coupling between the DPD RBC membrane and the DPD fluid is achieved as follows: First, the viscous interaction between a fluid DPD particle and a membrane DPD particle is adjusted to enforce the no-slip boundary condition. Second, a bounce-back algorithm of the fluid DPD particle from the membrane mesh is used to enforce the no-penetration boundary condition, which is a computationally expensive component in this algorithm. Alternatively, a strong repulsive force such as the Morse potential is used to avoid the penetration of fluid into the cell. Recently, a new immersed boundary-type approach was introduced in smoothed dissipative particle dynamics to achieve better accuracy of fluid-structure interactions (Li et al. 2017).

5. EXAMPLES AND FINDINGS

We provide a brief survey of squeezing fluid mechanics of a vesicle in a contracting microfluidic channel in Section 5.1 and an RBC during splenic filtration in Section 5.2. In Sections 5.3 and 5.4 we discuss examples of leukocytes squeezing through narrow constrictions.

5.1. Vesicle as a Biomimetic Model

The hydrodynamics of vesicles under confinement has been extensively studied as a biomimetic cell navigating through narrow capillaries with diameters in the micrometer range under pressure differentials (Abkarian & Viallat 2005, Pak et al. 2015, Agarwal & Biros 2020). Incorporating water permeability into a vesicle model has provided insights into the behavior of cells under strong confinement (Quaife et al. 2021). Semipermeable vesicles experience water influx or efflux under strong confinement: Numerical simulations reveal that vesicles with moderate membrane permeability (10^{-12} m²s/kg, 10 times smaller than the water permeability constant for a membrane with aquaporins) traverse more efficiently and can restore their water content within a fraction of a millisecond postpassage.

A vesicle membrane can be endowed with multiple lipid species as a model for cells in both experiments (Tian et al. 2007) and modeling (Zhao & Du 2011). For a multicomponent vesicle in a tapered narrow constriction, numerical simulations uncover that the reversal of the pressure gradient in the lubrication flow coincides with the boundaries of lipid domains

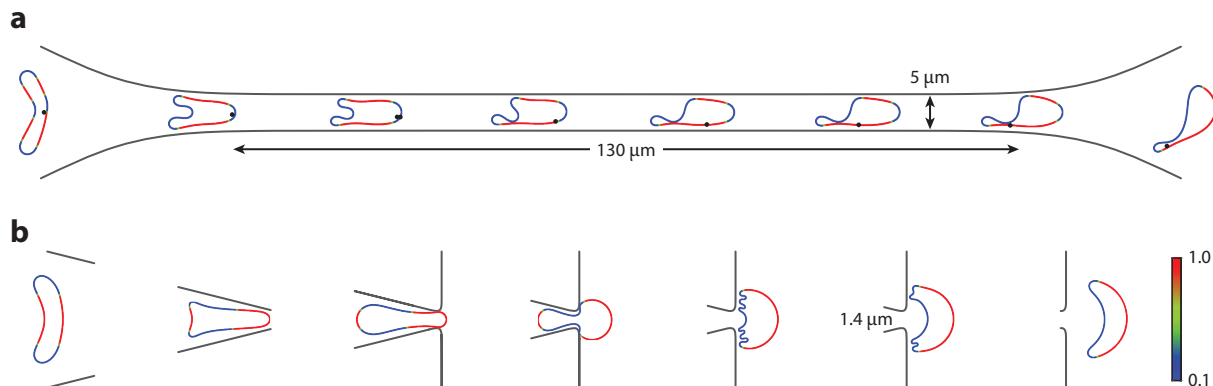


Figure 4

A two-component vesicle, composed of stiff (*red*) and floppy (*blue*) membrane domains, navigates two narrow constrictions. Lipid phase separation is arrested when the vesicle enters the stenosis (*a*) and the neck (*b*). Upon exiting, the two lipid species fully segregate, with the stiff domain leading and the soft domain trailing. Figure adapted from Gannon et al. (2024).

(Gannon et al. 2024). Furthermore, it is found that the lipid phase separation is arrested when the vesicle is in the stenosis (see **Figure 4a**). As soon as the vesicle moves past the constriction to the opening, the two lipid species fully separate with a domain of stiff lipids sorted to the front and a domain of floppy lipids sorted to the back (right side of **Figure 4**), resulting in an asymmetry in lipid species and a nonuniform tension distribution across the membrane, with lower tension at the front and higher tension at the back (Gannon et al. 2024). Such behavior mirrors that of RBCs traversing narrow channels, where negative membrane tension leads to rear-dominated compression (Secomb et al. 1986, Secomb 2017).

5.2. Splenic Filtration and Bone Marrow Egress of Red Cells: Geometry and Local Structural/Mechanical Transient Changes

Splenic filtration and bone marrow egress involve RBC movement through complex environments where their shape and flexibility are crucial. In the spleen, RBCs must deform to pass through narrow slits in the red pulp; only healthy cells reenter circulation, while damaged or aged ones are trapped and destroyed. The spleen's structure, including slit size and macrophages, facilitates this selective process. Similarly, the release of RBCs from the bone marrow depends on their ability to deform and pass through sinusoidal vessels. Reticulocytes must become more flexible as they mature, ensuring smooth egress into the bloodstream. Both the spleen and bone marrow serve as mechanical checkpoints, allowing only structurally sound RBCs to circulate.

To investigate this problem, Freund (2013) used 2D BI simulations to study the flow of RBCs through a narrow spleen-like slit and showed that simulated RBCs deform significantly in order to pass through small slits in the spleen, a process that may help remove aging cells from circulation as their cytosol viscosity increases with age. Simulations show that different flow rates and viscosities result in three behaviors: cell jamming at low flow, slow passage at higher viscosities, and infolding at high flow or viscosity, which may promote cell degradation. Pivkin et al. (2016) employed DPD to simulate the same problem and showed that the spleen selects RBCs based on their geometry and surface area, with implications for understanding diseases like malaria, hereditary spherocytosis, and thalassemia by providing biophysical limits for RBC passage through the IESs. Their analysis established critical bounds between surface area and volume for healthy RBCs, identifying thresholds beyond which RBCs fail the physical fitness test required to traverse the IES. These

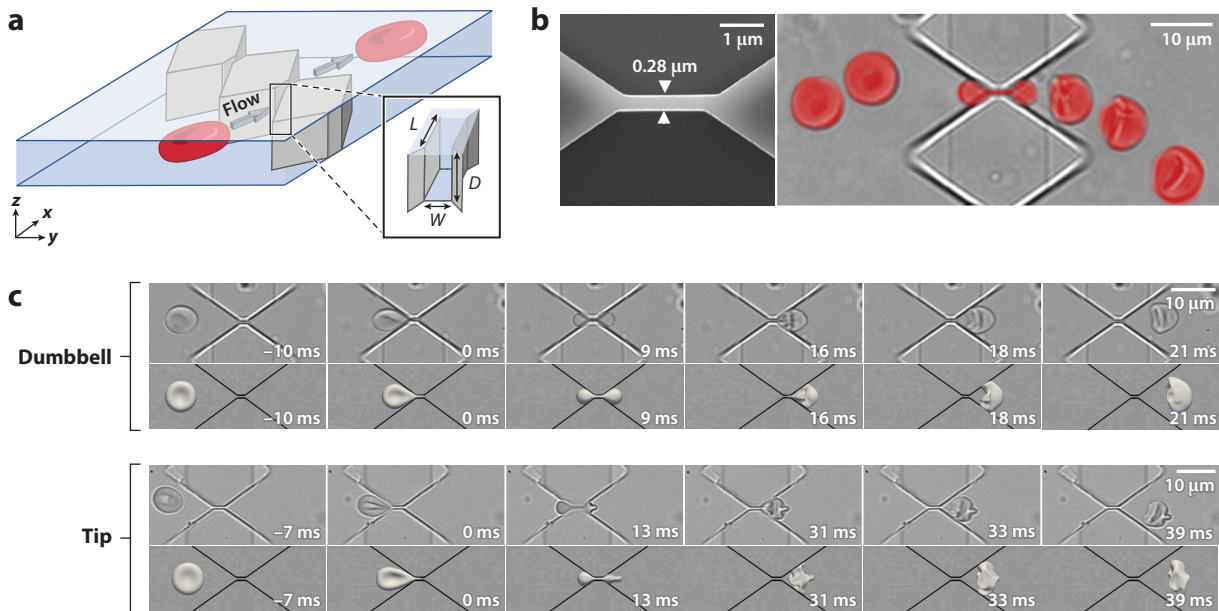


Figure 5

Illustration of the experiments and typical behaviors of red blood cells (RBCs) squeezing through submicrometer-wide slits. (a) Schematic of an RBC approaching a slit. The x , y , and z axes indicate the flow direction, width, and depth of the channel, respectively. (b) RBCs can pass through 0.28- μm -wide slits. (Left) Electron microscopy image of a 0.28- μm -wide silicon bridge (light gray) used for molding 0.28- μm -wide slits. (Right) Superimposed images of an RBC (red) passing through a $0.28 \times 1.87 \times 5.0 \mu\text{m}^3$ slit [bottom view in the (x, y) plane]. (c) Experimental observations (top rows) and numerical simulations (bottom rows) of squeezing RBCs displaying dumbbell and tip shapes at the slit exit, in $0.74 \times 2.12 \times 4.80 \mu\text{m}^3$ (top) and $0.86 \times 2.75 \times 4.70 \mu\text{m}^3$ (bottom) slits, respectively. Figure adapted from Moreau et al. (2023).

predictions show strong agreement with ex vivo experiments conducted on isolated perfused human spleens and synthetic spleen models based on microsphere filtration systems developed by Buffet and colleagues (Deplaine et al. 2011).

Recently, Moreau et al. (2023) used a combination of live imaging, microfluidic experiments, and multiscale modeling to investigate how RBCs pass through the narrow IESs of the spleen (Figure 5). They focused on the extreme RBC deformation as RBCs squeeze through slits as narrow as 0.28 μm and found that, to pass through these tiny slits, RBCs must adopt a specific geometry—two tether-connected spheres—and sometimes unfold their spectrin cytoskeleton (Figure 5b). Interestingly, the study revealed that the mechanosensitive PIEZO1 channel, often involved in cell mechanics, is not required for this process (see also Moreau et al. 2025). Instead, the viscosity of the RBC cytoplasm plays a crucial role in an RBC's passage through the slits. Experimental and computational results on the dependence of RBC transit time on applied pressures further demonstrated that the dynamic behavior of RBCs approximates that of a 2D Poiseuille flow of a Newtonian fluid, consistent with the scaling behavior described by Equation 4. Nonetheless, more detailed analyses (Borbas et al. 2023, Tang et al. 2023) indicate that surface tension and additional mechanical factors such as shear viscoelasticity, reflected in Equation 3, may also contribute to the observed dynamics as reported in similar simulations (Gürbüz et al. 2023). Finally, depending on the surface area-to-volume ratio, RBCs deform into dumbbell or tipping shapes (Figure 5c). Moreau et al.'s (2023) work also highlights the physiological relevance of splenic filtration, which ensures that damaged or diseased RBCs are removed from circulation.

Moreau et al. (2023) demonstrated that RBCs can traverse rigid slits as narrow as $0.28\ \mu\text{m}$ at body temperature. However, physiological IESs may exhibit limited flexibility, even though endothelial cells are substantially stiffer than RBCs. Li et al. (2024) employed a DPD model to investigate RBC passage through deformable IESs in the spleen, showing that incorporating slit deformability yields more accurate predictions of the critical surface area-to-volume ratio that governs the splenic clearance of senescent RBCs, compared to rigid-wall models.

5.3. Leukocytes Passing Through Pulmonary Capillary Beds

WBCs face significant mechanical challenges navigating the pulmonary microcirculation where capillary diameters range from 2 to $15\ \mu\text{m}$ (Doerschuk 2001) (see **Figure 6a**). Neutrophils ($10\text{--}15\ \mu\text{m}$) and monocytes ($15\text{--}30\ \mu\text{m}$) undergo successive squeezing through $40\text{--}100$ alveolar capillaries, where a 30-Pa pressure drop across each segment slows their transit, increasing local WBC concentration 50-fold. The slow strain rate experienced by WBCs leads to their temporary pauses at capillary entrances and hop-like motion, lagging behind plasma and RBCs (Doerschuk et al. 1993). Increased WBC rigidity can result in capillary entrapment, contributing to lung injury and ischemia, as seen in ARDS and sepsis. However, no single model fully captures WBC mechanical responses across different timescales, with existing models only addressing specific applied stresses (Herant et al. 2003, Shirai 2008).

Microfluidic studies have primarily examined the entry and movement of individual neutrophils and monocytes through narrow constrictions or channels (Yap & Kamm 2005, Preira et al. 2015). Cells either pass quickly or become trapped for an extended duration (Gabriele et al.

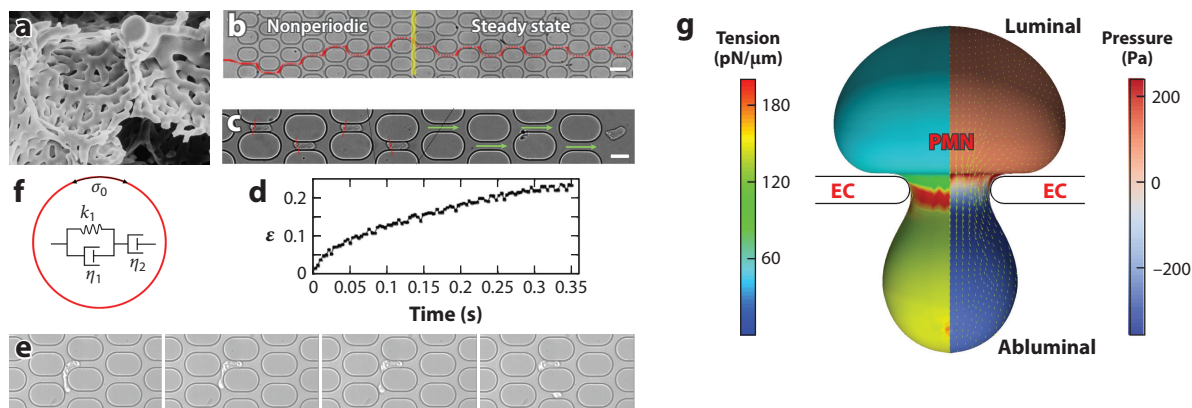


Figure 6

Studies of leukocytes passage in (a–f) the pulmonary capillary bed and (g) diapedesis. (a) Scanning electron microscopy of a vascular cast prepared from a rat specimen. (b) Nonperiodic trajectory before reaching a steady-state regime. The red dots correspond to the position of the cell center of mass. Scale bar: $20\ \mu\text{m}$. (c) Asymmetric motion in the channels and surprising left turn after exiting the network. Superimposed images show a cell passing through the network. Green arrows show the flow velocity field. Scale bar: $15\ \mu\text{m}$. (d) Elongation ratio ε of a cell while it enters the first channel. (e) A Lat cell with a high capillary number splitting into two parts like a viscous drop. (f) Cell rheological model: a viscoelastic fluid combining a Kelvin–Voigt solid (η_1 , k_1) in series with a dashpot η_2 , and with a cortical tension σ_0 . (g) Cross-section view of coupled finite element and boundary element simulations of a polymorphonuclear neutrophil (PMN) passing through a pore between endothelial cells (ECs) under a pressure drop of $500\ \text{Pa}$. The left half and its color map show the tension on the neutrophil membrane with a unit of piconewtons per micrometer. The right half and its color map show the pressure on the neutrophil membrane. The arrows on the right side show the velocity field of the flow inside the neutrophil. The pore is $3\ \mu\text{m}$ in diameter and the neutrophil is $8.5\ \mu\text{m}$ in diameter. Panel a adapted with permission from Guntheroth et al. (1982). Panels b–d and f adapted from Dupire et al. (2020). Panel g adapted from Mukhopadhyay et al. (2024) (CC BY 4.0).

2010). Mechanical properties such as the loss modulus can be extracted from their transit (Preira et al. 2013). Whole blood from sepsis and trauma patients has also been analyzed for rheological properties (Nishino et al. 2005). Monocyte transport across a biomimetic pulmonary capillary network (Dupire et al. 2020) was studied using 50 successive constrictions (15 μm long, 10 μm wide, 9.3 μm high) under 10–40-mbar pressures (physiological range). Initially, monocytes displayed irregular movement, stopping and making random turns at bifurcations (**Figure 6b**). Deeper into the network, they deformed more, moved faster, and reached a steady state, alternating left and right turns periodically (**Figure 6c,d**). In contrast, giant lipid vesicles exhibited random trajectories, frequently hopping and rupturing under stress, without the gradual adaptation observed in monocytes.

The membrane mechanics of WBCs governs their dynamics and transport in the capillary network (Dupire et al. 2020), similar to RBCs squeezing through the splenic slit in Section 5.2. A WBC's cortical tension σ_0 , induced by the actin polymer network beneath the lipid-bilayer membrane and characterized by the capillary number $\text{Ca} = \eta_e v_{\text{fluid}} / \sigma_0$ (for a characteristic fluid velocity v_{fluid}), is found to be directly related to its dynamic and transport: WBCs with a high Ca (Lat cells with actin polymerization inhibited and rigidity reduced by latrunculin A) quickly reached a steady state with a rapid transit, minimal arrest, and occasional division at bifurcations (**Figure 6e**). WBCs with a low Ca (Jasp cells with actin polymerization promoted and rigidity enhanced by jasplakinolide) moved slowly with irregular stops at bifurcations, and only 46% reached a steady state.

Monocyte behavior in the biomimetic network has two key characteristics (Dupire et al. 2020): (a) a rapid, small viscoelastic deformation over tens of milliseconds and (b) a slower, 10% viscous deformation over hundreds of milliseconds. Based on these two dynamic features, Dupire et al. (2020) modeled the monocyte as a viscous element ($\eta_2 = 45 \text{ Pa}\cdot\text{s}$) in combination with a viscoelastic Kelvin–Voigt body, with an elastic modulus of $k_1 = 1,000 \text{ Pa}$ and a viscosity of $\eta_1 = 10 \text{ Pa}\cdot\text{s}$, and a cortical tension of σ_0 (see **Figure 6f**).

A monocyte rapidly adapts its shape when it resides in the horizontal channel for tens of milliseconds. In the diagonal channel, a monocyte's residence time is of the order of hundreds of milliseconds, allowing an additional 10% viscous deformation, which is asymmetrical due to the channel geometry and not fully relaxed in the next horizontal channel, where its residence time is shorter. Consequently, asymmetrical deformation is maintained in the horizontal channel, enabling the cell to turn immediately to the right or left as it exits the channel and easily enter the next diagonal channel. Optimum steady-state cell deformation was gradually achieved by alternating fast and slow cell transit in successive meshes.

Finally, once in the steady state (**Figure 6b**), balancing the hydrodynamic force pushing the monocyte with the frictional force between the cell membrane and the channel wall allows an experimental determination of the viscous coefficient k_f (due to the shear force inside the lubrication layer), the lubrication layer thickness $\delta = \eta_e / k_f$, where η_e is the viscosity of the external fluid, and the tension at the front of the cell σ_f using the results from Bretherton (1961). For Lat cells (high Ca), $k_f = 2.4 \text{ mPa}\cdot\text{s}/\text{m}$, $\delta = 410 \text{ nm}$, and $\sigma_f = 210^{-6} \text{ N}/\text{m}$. For Jasp cells (low Ca), $k_f = 7.6 \text{ mPa}\cdot\text{s}/\text{m}$, $\delta = 130 \text{ nm}$, and $\sigma_f = 10^{-3} \text{ N}/\text{m}$.

This study by Dupire et al. (2020) showed how the high internal viscosity [likely due to the nucleus as in the case of a compound vesicle (Veerapaneni et al. 2011)] of monocytes allowed their gradual asymmetric shape evolution, enabling their efficient motion without arrests in the pulmonary bed, in contrast to low-viscosity giant vesicles. Noteworthy, no active process was required to describe this adaptive behavior.

5.4. Leukocyte Diapedesis

Neutrophil diapedesis, a critical step in the immune response, involves a complex series of mechanical and molecular events as neutrophils exit the bloodstream and move through the endothelial cell layer into tissues (Muller 2011). Although the pressure difference between the blood vessel lumen and the tissue space might play a role in pushing the neutrophil through the gap (**Figure 1e,f**), unlike RBCs, neutrophils mainly move by themselves using internal active actomyosin structures following the chemokine gradients caused by inflammation or infections.

Under fluid shear stress in the blood flow, neutrophils undergo margination and rolling along the vessel walls as a preparatory step for diapedesis (Freund 2007, Kumar & Graham 2012a). Meanwhile, blood flow also creates chemokine gradients (e.g., CXCL8) that guide neutrophil activation and adhesion (Cambier et al. 2023). Once firmly adhered to the endothelial cells to withstand the fluid shear forces (Isfahani & Freund 2012), neutrophils then employ podosomes—actin-rich protrusions—that help probe for weakened areas in the endothelial layer (Schwartz et al. 2021). During the transendothelial migration, neutrophils induce mechanical forces to move through endothelial junctions, creating physical openings for diapedesis (Schwartz et al. 2021). Given that the cytoplasmic viscosity of neutrophils is estimated to be five orders of magnitude higher ($\sim 10^2$ Pa·s) than that of RBCs, the associated transit times are correspondingly slower—on the order of 10 min—compared to the approximately 10 ms required for RBCs to pass through splenic slits, as shown in **Figure 5c** (Moreau et al. 2025).

Vascular endothelial cells (VECs) actively facilitate neutrophil transmigration by engulfing and pulling neutrophils through endothelial junctions, a biomechanically coordinated remodeling process. Neutrophil binding triggers signaling pathways that enhance endothelial contractility, increasing cytosolic calcium and myosin activity to expand junctions via F-actin tension and VE-cadherin modulation. Neutrophils then generate vertical and tangential forces, using their stiff nuclei to push through. VECs also form a transmigratory docking structure, a microvilli-like projection anchoring neutrophils through F-actin assembly and contractile signaling (Heemskerk et al. 2016). This interplay enables efficient transmigration while maintaining junctional integrity. Yeh et al. (2018) measured the spatial and temporal evolution of 3D traction stresses generated by leukocytes and VECs to understand the mechanical processes involved in paracellular diapedesis. Their findings suggest that contractile stresses from both leukocytes and VECs independently disrupt junctional tension, initiating gap formation before transmigration.

Mukhopadhyay et al. (2024) used genetic, imaging, microfluidic, and in vivo methods, as well as computational modeling, to show that the mechanosensor PIEZO1 in polymorphonuclear neutrophil (PMN) membranes triggers spike-like Ca^{2+} signals. Based on the membrane tension, PIEZO1 activation probability is predicted, linking its mechanosensing role to enhanced PMN bactericidal function in tissue. Mice lacking PIEZO1 in PMNs exhibited impaired bacterial clearance and increased susceptibility to severe lung infections. This finding is consistent with those that demonstrated the roles of PIEZO1 in leukocyte diapedesis (Friedrich et al. 2019, Zhong et al. 2020, Wang et al. 2022). Furthermore, by combining the BI method and the finite element method, both the maximum membrane tension near the pore and the internal flow are quantified during the diapedesis (see **Figure 6g**). In addition, pore migration assays—such as Transwell chambers (Rumianek & Greaves 2020) and microfluidic platforms (Boneschansker et al. 2014, Mukhopadhyay et al. 2024)—serve as valuable tools for investigating leukocyte migration, chemotaxis, and transmigration. These systems have elucidated how chemokine gradients, cytoskeletal remodeling, and nuclear stiffness collectively regulate leukocyte movement under confinement.

FUTURE ISSUES

1. The dynamic remodeling of a lipid-bilayer membrane undergoing topological changes can be incorporated in a continuum model suitable for boundary integral formulation. Giant vesicles can be reconstituted with both channel proteins and actomyosin networks to maintain mechanical integrity when they are squeezed through strong confinement.
2. Molecular-scale interactions can be integrated between mechanosensitive channels, cytoskeletal architecture, and lipid bilayer into modeling red blood cells (RBCs) to address issues of RBC aging. Experimental techniques for fast and accurate detection of cell volume change, PIEZO1 activation, and ATP release as RBCs squeeze through submicrometer openings can be developed. It is important to understand the mechanisms of nanovesicle release from highly deformed RBCs, as these vesicles can serve as diagnostic biomarkers and drug delivery cargoes.
3. Models can be refined to study more complicated leukocyte diapedesis and cancer extravasation by incorporating cytoskeletal dynamics, cytosol flow, adhesion dynamics, membrane mechanics, and signaling pathways and to compare these with experiments.

DISCLOSURE STATEMENT

The authors are not aware of any affiliations, memberships, funding, or financial holdings that might be perceived as affecting the objectivity of this review.

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